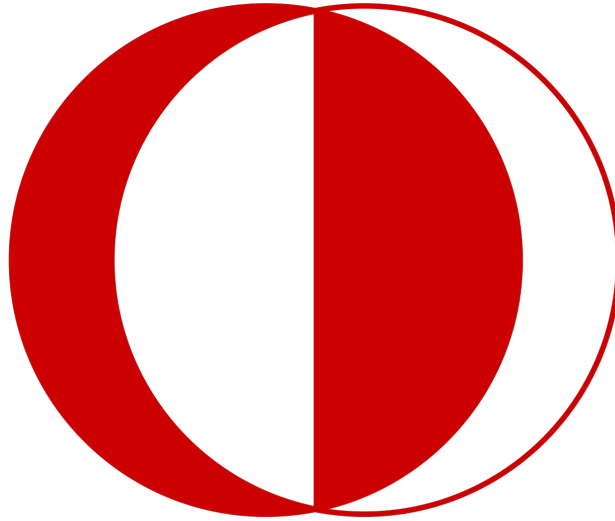


MIDDLE EAST TECHNICAL UNIVERSITY

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Department of Statistics



COMPARATIVE ANALYSIS OF GLP-1 RECEPTOR AGONISTS

Patient Characteristics and Adverse Events Profiles

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Introduction

In recent years, medications such as Ozempic, Trulicity, and Mounjaro have gained significant worldwide popularity for regulating blood sugar levels and managing weight. To understand how these medications affect and their safety profile, it is important to analyze how the human body processes food.

GLP-1 (Glucagon-Like Peptide-1) is released by the human intestines whenever food is consumed. The primary role of GLP-1 is to send a message to the brain and digestive system that the body has received food and is fulfilled. However, naturally produced GLP-1 is highly delicate; the body's enzymes degrade and eliminate it in a couple of minutes (Müller et al., 2019).

Ozempic, Trulicity, and Mounjaro are the medications that are classified as GLP-1 receptor agonists. The body's organs work as a system of locks, and hormones as the keys. The GLP-1 hormone, which is naturally released by the intestines, is a key that unlocks but breaks afterwards. On the other hand, GLP-1 agonists are durable and scientifically created, which can be replicated by the natural hormone. They remain in the lock and keep the system active for days, because they are resistant to the body's breakdown system (Collins & Costello, 2024).

Three major changes appear in the body when the GLP-1 receptors are activated by these medications:

- **In the Brain (Hunger Control):** The hunger centers are stimulated by these medications, which leads to the development of a consistent feeling of satisfaction and reduces food cravings.
- **In the Stomach (The Speed of Digestion):** The transition of the food into the intestines is slowed down by the medications. Since the food stays in the stomach much longer, patients experience fullness even though they ate only a small portion hours ago.
- **In the Pancreas (Blood Sugar Levels):** The pancreas is helped by the medication to release insulin in response to high blood sugar levels, which allows it to regulate blood glucose safely without causing dangerous declines.

This change in physiological state results in adverse effects, even though these medications function effectively in the system management. Nausea and other gastrointestinal problems are the body's first response to adapt to the new, slowed digestion process. As the number of patients who use these medications increases, understanding how the clinical complications and demographic factors affect the patient experience becomes important.

This statistical report aims to provide the real-world adverse event profiles of these medications. By using 2025 Quarterly Datasets from the United States Food and Drug Administration’s (FDA) Adverse Event Reporting System (FAERS; 2025), patient demographics, which include age and weight distributions of the patients on the specific GLP-1 medications, specific side effects, which cover the occurrence rate of nausea, and overall clinical complexity, which determines the total quantity of adverse events a patient experiences.

By transforming complicated, real-world health data into clear observations about the tolerability of the most popular GLP-1 medications available through an extensive statistical analysis, this research aligns with global adjectives. Specifically, this analysis is relevant to SDG 3 (Good Health and Well-being), as adverse event monitoring is important for improving health outcomes and medication safety in disease therapy. It also supports SDG 5 (Gender Equality), as the analysis contains a comparison between female and male patients, which promotes gender-sensitive health monitoring. Finally, it relates to SDG 10 (Reduced Inequalities) since comparing variances among different patients aids in identifying and addressing potential disparities in healthcare experiences.

Data Preprocessing

Data Source

The FDA Adverse Event Reporting System (FAERS), a publicly accessible pharmacovigilance database run by the U.S. Food and Drug Administration (FDA), provided the data used in this project. FAERS gathers unplanned reports of adverse medication events from manufacturers, consumers, and medical experts. Quarterly datasets from 2025 Q1–Q4 were downloaded and merged.

Dataset overview:

Table 1. Dataset overview

Dataset	Rows	Purpose
DEMO	1.6M	Patient Demographics
DRUG	7.8M	Drug Exposure
REAC	5.5M	Adverse Events
FINAL TABLE	27771	

Duplicate reports, inconsistent drug names, missing demographic data, several suspect drugs per case, and varied dose forms were all included in the raw FAERS files. As a result, thorough preprocessing and filtering were necessary before statistical analysis. DuckDB (Python Library) was used to combine

and clean the quarterly FAERS datasets from 2025 in one final table. Only principal suspect (PS) medicines and single-PS cases were kept to minimize confusion. Ozempic, Trulicity, and Mounjaro reports were included in the final dataset.

DEMO (Patient Demographics) Preprocessing:

Table 2. DEMO Preprocessing

Application	Explanation
Quarter Merge	Q1-Q4 are combined
Age Conversion	MON/WK/DY is converted to years
Weight Conversion	LB to KG
Invalid Ages Removed	Removed ages > 120
Invalid Weights Removed	Removed weights > 300
Sex Filtering	Only M/F

Final columns in DEMO are:

primaryid, age_years, weight_kg, sex, country

DRUG Preprocessing:

Table 3. DRUG Preprocessing

Application	Explanation
Only PS Drugs	Primary Suspect Only
Single PS Filtering	Removed Multiple PS Cases
Target Drugs Selected	Ozempic, Trulicity, Mounjaro
Dose Standardization	Converted to mg
Concomitant Drugs Counted	Polypharmacy Variable

Final columns in Drug are:

primaryid, drugname, dose_mg, num_concomitant_drugs

REAC Preprocessing:

Table 4. REAC Preprocessing

Application	Explanation
PT Cleaning	Standardized Preferred Terms
Duplicate Reactions Removed	DISTINCT
Binary Outcomes Created	nausea_binary etc
num_reactions Created	Total Event Count

Final columns in Reac are:

primaryid, pt, nausea_binary, vomiting_binary, diarrhea_binary, num_reactions

Final Table:

DEMO, DRUG, and REAC are assembled by the primaryid column. Final case number is 27771.

Table 5. Final Table

Variable	Type
drugname	categorical
age_years	continuous
weight_kg	continuous
num_reactions	continuous
num_concomitant_drugs	continuous
nausea_binary	binary

Exploratory Data Analysis

To analyze the final FAERS dataset's structure and determine whether the chosen variables were suitable for the intended statistical tests, exploratory data analysis was carried out. The final case-level dataset contained reports for three chosen GLP-1 receptor agonists: Mounjaro, Ozempic, and Trulicity, following the merging and cleansing of the 2025 quarterly FAERS files. To minimize uncertainty in drug-event attribution, only primary suspect substances and single-primary-suspect instances were kept. All the observations in the dataset are assumed independent because each row of different variables in the dataset

is linked to a unique patient, which is identified by a unique “primaryid” or “caseid”. This means that there are no repeated measures for the same individuals.

Reports were not distributed equally among the three medications. The highest number of instances came from Mounjaro, which was followed by Ozempic and Trulicity. Since subsequent tests evaluate adverse event outcomes across treatment groups, it was crucial to find this imbalance before formal analysis. Because age was included in both the one-sample and two-sample mean analyses, it was investigated. Although there was some variance in dispersion, the age distributions of the chosen medications were generally similar. This validated the use of age as a continuous variable in the regression model and hypothesis testing. Also, GLP-1 receptor agonists are intimately associated with metabolic illness and weight control; body weight was also investigated. The weight distributions demonstrated sufficient variance among drug groups to support their inclusion in the multiple regression model and ANOVA.

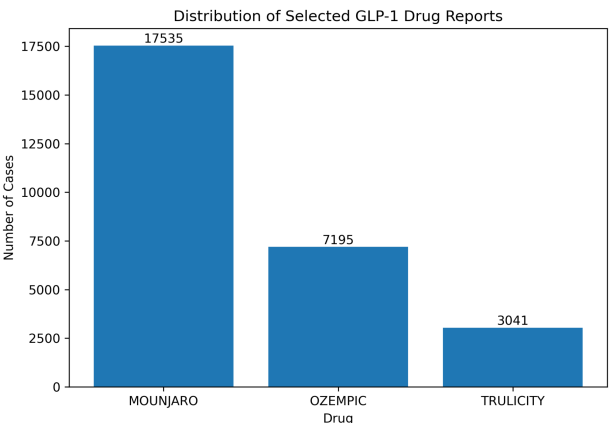


Figure 1. Distribution of Selected GLP-1 Drug Reports

The most commonly reported preferred terms were used to analyze adverse event patterns. Among the most frequently reported effects were gastrointestinal problems, such as nausea, vomiting, and diarrhea. The choice to utilize nausea as the primary binary adverse event measure in the proportion-based analyses was supported by this trend.

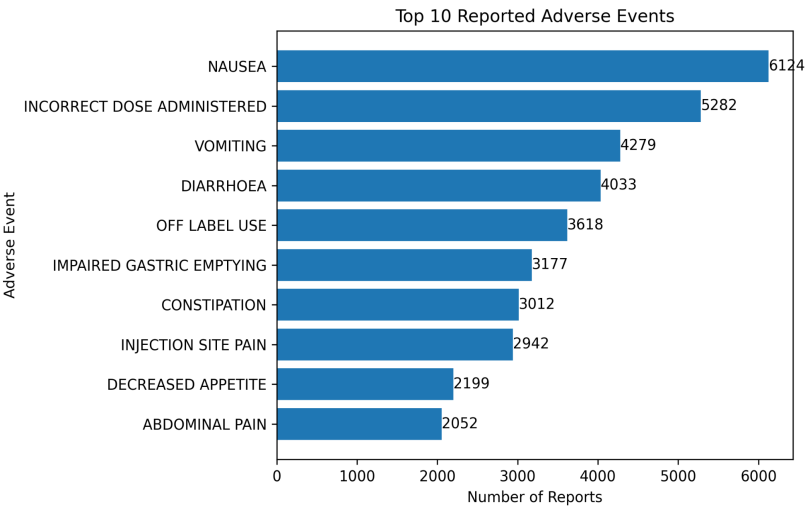


Figure 2. Top 10 Reported Adverse Events

- Is the mean age of Ozempic users significantly different from 60 years?

$H_0: \mu = 60$ (The mean age of Ozempic users is equal to 60 years.)

$H_A: \mu \neq 60$ (The mean age of Ozempic users is significantly different from 60 years.)

Before performing the one-sample t-test, relevant assumptions were verified. Although the distribution was not perfectly normal and exhibited minor deviations in the extreme tails, the Ozempic group had a large sample size ($n = 7,195$). According to the Central Limit Theorem, the one-sample t-test is resilient to these normality deviations. In addition, a sensitivity analysis that included only adult patients aged 18 and older reached the same statistical conclusion, indicating that the result is consistent.

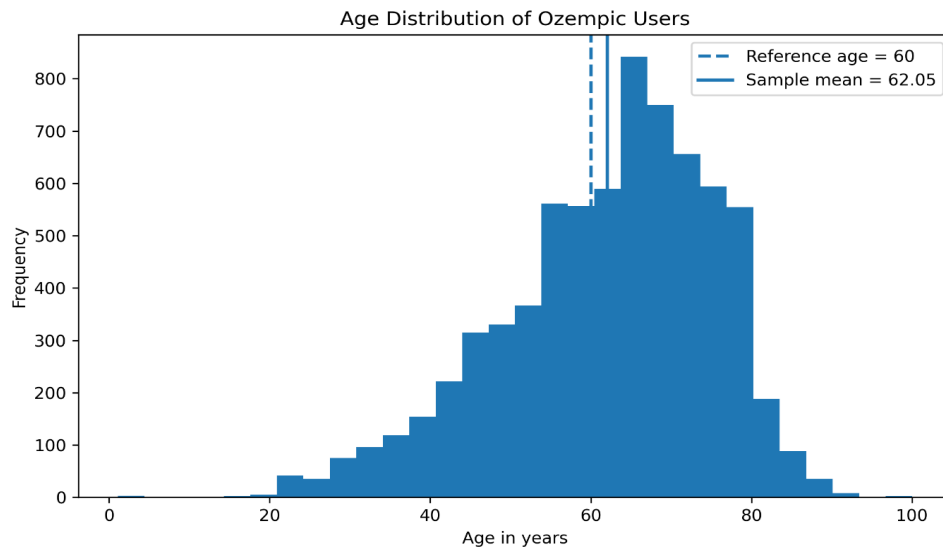


Figure 3. Age distribution of Ozempic users

The histogram depicts the age distribution of Ozempic users. A dotted vertical line indicates the 60-year reference value, whereas a solid vertical line reflects the sample mean. The visual indicates that the sample mean is higher than 60, with the majority of the patient population falling in the senior age groups. The null hypothesis is rejected, and the average age of Ozempic participants is significantly greater than 60 years.

- **Is there a significant difference in the average age of patients using Ozempic compared to those using Trulicity?**

$$H_0: \mu_{Ozempic} = \mu_{Trulicity}$$

$$H_A: \mu_{Ozempic} \neq \mu_{Trulicity}$$

Before applying the test, all assumptions must be checked:

The assumption of theoretical normality is rejected by the Shapiro-Wilk test, which reveals a significant deviation from normality ($p < 0.05$). However, the Central Limit Theorem guarantees a normal distribution of sample means in a large sample, even though real data depart from normality. Two samples have uneven variances, which is demonstrated by Levene's test. Because of this, Welch's two-sample t-test is the appropriate method to use for comparing their sample means.

A two-sample Welch's t-test is conducted to compare the average age of patients who are Ozempic users and Trulicity users. The age distributions of Ozempic users and Trulicity users are significantly different ($p < 0.001$). Moreover, Trulicity users have a higher sample mean age than the Ozempic users due to the boxplot that can be seen in Figure X, which indicates an upward shift in medians and interquartile range for Trulicity compared to Ozempic. The null hypothesis was rejected ($p < 0.001$). There is significant statistical evidence to support that the average age of the groups varies considerably, with users of Trulicity differs on mean than users of Ozempic.

- **Is the proportion of patients experiencing nausea significantly different from 20% among Ozempic users?**

$$H_0: p = 0.20$$

$$H_A: p \neq 0.20$$

Before applying the test, all assumptions must be checked:

The expected number of successes ($n \cdot p_0$) and the expected number of failures ($n \cdot (1 - p_0)$) must be at least 10 for the normal approximation to be valid. Since the expected number of successes is 1439 and the expected number of failures is 5756, the one-sample proportion test can be applied.

A one-sample proportion test with continuity correction is applied to determine if the proportion of nausea among Ozempic users differs significantly from the hypothesized value, which is 20%. The assumptions for the normal approximation are satisfied since there are at least 10 successes and 10 failures.

The 19.67% of the patients experienced nausea ($\hat{p} = 0.1967$). The test statistics X – squared is 0.4792 with 1 degree of freedom, and the p-value is 0.4886. The null hypothesis is rejected since the p-value (0.4886) is greater than the standard significance level (0.05). There is not enough statistical evidence to state that the percentage of patients using Ozempic who experience nausea is significantly different from 20%.

- **Is there a significant difference in the proportion of patients experiencing nausea between Ozempic and Trulicity users?**

$$H_0: p_{\text{Ozempic}} = p_{\text{Trulicity}}$$

$$H_A: p_{\text{Ozempic}} \neq p_{\text{Trulicity}}$$

Before applying the test, all assumptions must be checked:

For each group, the expected number of successes ($n \cdot p_0$) and the expected number of failures ($n \cdot (1 - p_0)$) must be at least 10 for the normal approximation to be valid. For the Ozempic group, the expected number of successes is 1415, and the expected number of failures is 5780. For the Trulicity group, the expected number of successes is 644, and the expected number of failures is 2397. So, the two-sample proportion test is applicable.

A two-sample proportion test with continuity correction is applied to compare the nausea rates of patients using Ozempic and those using Trulicity. The assumptions for the normal approximation are satisfied since there are at least 10 successes and 10 failures for both groups. The sample proportions of patients experiencing nausea for Ozempic users is 19.67% ($\hat{p}_1 = 0.1967$) and for Trulicity users is 21.28% ($\hat{p}_2 = 0.2128$). The test statistic X-squared is 2.943 with 1 degree of freedom, and the p-value is 0.08625. Furthermore, the 95% confidence interval for the difference in populations is $[-0.0325, 0.0023]$.

Figure 4 represents a mosaic plot visualizing the cross-tabulation of drug type (Ozempic vs. Trulicity) and the occurrence of nausea. In this plot, the width of the columns demonstrates the total number of patients taking the medications. It can be seen that the sample size of Ozempic is larger than the sample size of Trulicity. The height of the columns represents the proportion of patients who experienced nausea. It can be observed that the line that divides “Nausea” and “No Nausea” is at nearly the same level for both Ozempic and Trulicity groups. This alignment verifies the statistical findings from the two-sample proportion test. This shows that the rate of nausea is independent of the treatment group. The null hypothesis is rejected since the p-value (0.08625) is greater than the standard significance level,

which is 0.05. There is not enough statistical evidence to state that the rate of nausea differs between Ozempic and Trulicity users.

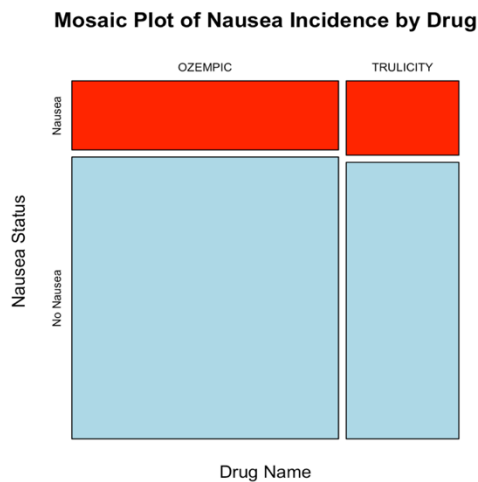


Figure 4. Mosaic Plot of Nausea Incidence by Drug

- **What are the independent effects of drug type (Mounjaro, Trulicity, Ozempic), age, weight, and the number of concomitant drugs on the frequency of reported adverse events?**

H₀: The frequency of reported adverse events (log-transformed) is unaffected by the specified independent variables.

H_A: The frequency of reported adverse events (log-transformed) is significantly affected by at least one of the independent variables.

A multiple linear regression model is fitted to explain the independent impacts of drug type, age, body weight, and the number of concomitant drugs on the log-transformed number of reported adverse events.

The overall model is statistically significant: $F(5, 5240) = 191.0$ $p < 0.001$ with an R squared value of: $R^2 = 0.154$, which means that the model explained 15.4% of the variability in the number of reported adverse events. Clinical data usually contains low R^2 values since individual differences are not measured. According to Wooldridge, generating unbiased estimators of the effects of the independent factors is not directly impacted by the size of the R^2 . In addition, Shmueli (2010) notes that explanatory power and predictive ability should not be confused. The aim of explanatory modelling is to measure the power of casual connections rather than to predict future data. The cases of individuals who are using Ozempic were associated with a higher number of reported adverse events, compared to Mounjaro cases:

$\beta = 0.171$, $p < 0.001$. Trulicity cases are showing an even larger positive association: $\beta = 0.393$, $p < 0.001$. The number of concomitant drugs is positively associated with the number of adverse events: $\beta = 0.020$, $p < 0.001$, showing that adverse events were more frequently reported in cases involving more medicines. There is a small but statistically significant negative association between the number of adverse events and the ages of individuals: $\beta = -0.002$, $p < 0.001$. However, body weights of the individuals are not associated with the reported adverse event counts. $p = 0.864$. These extremely important p-values clearly explain the reliability of the direct connection between drug types and adverse events regardless of the low R^2 value. This figure shows the regression coefficients and corresponding 95% confidence intervals for all predictors included in the model.

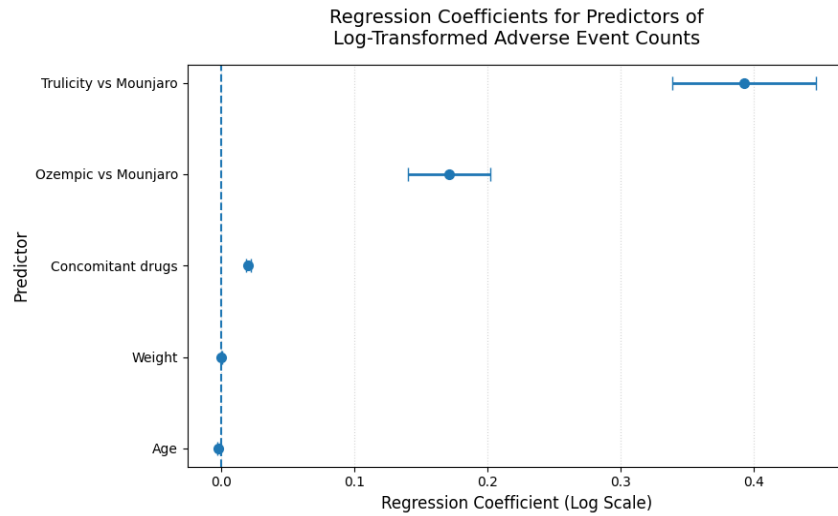


Figure 5. Regression Coefficients for Predictors of Log-Transformed Adverse Event Counts

- Do the mean weights of patients differ among Mounjaro, Ozempic, and Trulicity users?

$$H_0: \mu_{Mounjaro} = \mu_{Ozempic} = \mu_{Trulicity}$$

H_A : At least one drug group has a different mean patient weight.

Note: The analysis only includes patients with non-missing weight_kg values. Although the dataset had 27,771 observations, only 5,246 patients had available weight information

Trulicity customers had the highest mean patient weight, with an average of 108.80 kg. Mounjaro users have a mean weight of about 96.30 kg, whereas Ozempic users have the lowest mean weight, around

94.92 kg. The median values follow a similar trend, with Trulicity having the highest median weight. Trulicity also had the highest standard deviation, indicating more fluctuation in patient weights in this category.

The residual normality test and Levene's test for equality of variances were both significant. This indicates that the normality and equal variance assumptions are not completely satisfied. However, because the sample size is substantial, one-way ANOVA can still be used as an approximation, albeit the findings should be regarded with caution.

Table 6. One-way ANOVA results for patient weight by drug group

Source	df	Sum Sq	Mean Sq	F-value	p-value
Drug name	2	75,383.265	37,691.633	44.629	< 0.001
Residual	5,243	4,427,982.289	844.551		

The one-way ANOVA result is statistically significant: $F(2,5243) = 44.629$, $p < 0.001$. Thus, the null hypothesis is rejected. This indicates that the mean patient weight varies between the three medication groups. At least one of the Mounjaro, Ozempic, and Trulicity users has a considerably different average weight. However, ANOVA merely indicates that there is a difference between the groups. It does not specify which medication groupings differ from one another. Thus, Tukey multiple comparisons were used.

Table 7. Tukey multiple comparisons for patient weight by drug group

Comparison	Mean difference	p-value	95% CI	Result
Ozempic – Mounjaro	-1.371	2.329	[-3.343, 0.600]	Not significant
Trulicity – Mounjaro	12.504	< 0.001	[9.055, 15.952]	Significant
Trulicity – Ozempic	13.875	< 0.001	[10.404, 17.345]	Significant

The Tukey test reveals that Trulicity users have a considerably greater mean weight than Mounjaro and Ozempic participants. The average weight difference between Trulicity and Mounjaro is approximately 12.50 kg, whereas the average difference between Trulicity and Ozempic is approximately 13.87 kg. Both differences are statistically significant. However, the difference between Mounjaro and Ozempic is not statistically significant, with a p-value of 0.2329, which is more than 0.05. This indicates that Mounjaro and Ozempic users have similar mean weights in this sample.

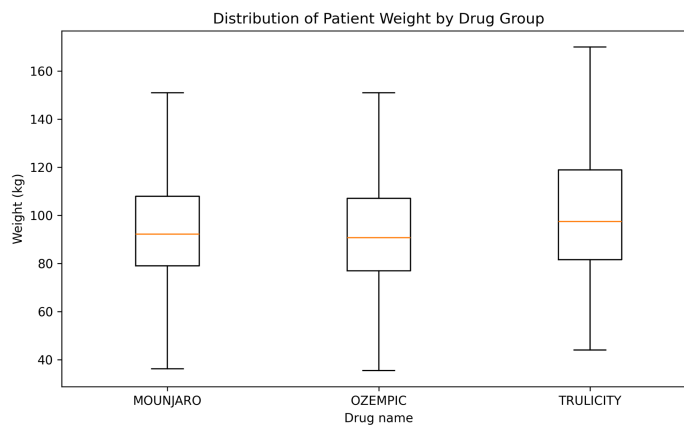


Figure 6. *Distribution of patient weight by drug group*

Trulicity users have larger weights and a broader distribution than the other two drug groups. Mounjaro and Ozempic distributions are more similar, with Mounjaro appearing somewhat higher on average. The greater spread in the Trulicity group confirms the Levene test result, which showed unequal variances among groups. The mean confidence interval plot demonstrates that Trulicity has a much greater mean patient weight than the other two medicines.

- **Does the number of reported adverse events differ by drug type and sex?**

Two-way ANOVA was applied to examine whether the mean number of reported adverse event reactions differs according to drug type, sex, and the interaction between drug type and sex.

H₀: The mean number of reported reactions is equal across all drug groups.

H_A: At least one drug group has a different mean number of reported reactions.

For the **sex effect**, the hypotheses are:

H₀: The mean number of reported reactions is equal for female and male patients.

H_A: The mean number of reported reactions differs between female and male patients.

For the **interaction effect**, the hypotheses are:

H₀: There is no interaction between drug type and sex.

H_A: The effect of sex on the mean number of reported reactions depends on the drug.

The model used in this analysis is:

`num_reactions ~ drugname + sex + drugname * sex`

The response variable, `num_reactions`, is a numerical count, whereas `drugname` and `sex` are categorical. The residual normality test and Levene's test both yielded significant results, showing that the normality and equal variance assumptions were not met. The boxplot also demonstrates the skewness of the values. Since the sample size is large enough, the ANOVA results can still be used as an approximation, but they should be interpreted cautiously. As a result, the data should be interpreted as disparities in reported reaction counts, rather than as causative evidence that one drug actually causes more adverse events.

Ozempic users have the highest mean number of reported responses among both male and female patients. For Ozempic, the average number of reported reactions is 4.175 for females and 3.749 for males. Trulicity has an average value of 3.223 for females and 2.668 for males. Mounjaro has the fewest recorded reactions, with an average of 2.410 for females and 2.168 for males. Female patients report more responses than male patients in all three drug categories. The ANOVA analysis indicates a statistically significant main effect of drug type on reported reactions, $F(2, 27765) = 1235.885$, $p < 0.001$, leading to the rejection of the null hypothesis regarding medication effects. This suggests substantial variation in reactions among users of Mounjaro, Ozempic, and Trulicity. Additionally, the main effect of sex is significant, $F(1, 27765) = 111.193$, $p < 0.001$, implying that male and female patients report different average numbers of reactions. Furthermore, the interaction effect between drug type and sex is significant as well, $F(2, 27765) = 6.834$, $p = 0.001$, indicating varying differences in reported reactions between genders across the different drugs. Specifically, the data reveal a smaller female-male difference for Mounjaro and a larger difference for Ozempic and Trulicity.

In brief, the boxplot indicates that Ozempic groups have higher reported reaction counts, whereas Mounjaro groupings have lower values. There appears to be truth between these two medications. Female groups are often slightly higher than male groups while using the same medicine.

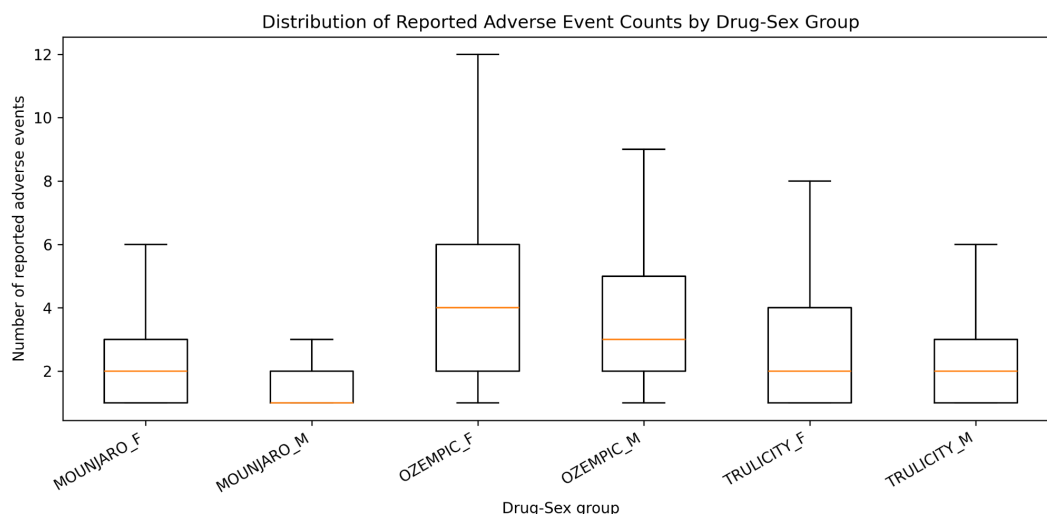


Figure 7. Distribution of Reported Adverse Events by Drug-Sex Group

Conclusion

This statistical analysis aims a comprehensive evaluation of real-world safety profiles and patient demographics of GLP-1 receptor agonists, which are Mounjaro, Ozempic, and Trulicity. 27,771 unique patient cases from the 2025 FDA Adverse Event Reporting System (FAERS) dataset are analyzed to conduct exploratory data analysis and inferential hypothesis testing to determine how the clinical and demographic factors influence adverse event reporting. There are several aspects to consider when evaluating safety profiles of these medications, which can be separated as three groups:

1. Demographic Variations (Age and Weight): The results suggest that the demographic profiles of patients do not have an exact stereotype across different GLP-1 treatments. Since the active ingredient in Ozempic is semaglutide, published studies of semaglutide treatment populations were observed. The mean age of patients close to 60 years. Therefore, the clinical reference baseline of 60 years was chosen, and results show that the mean age of patients who use Ozempic in the dataset is statistically higher than the reference. Furthermore, applied tests claim that patients who use Trulicity are older and have significantly higher mean body weight than both Ozempic and Mounjaro users. Since GLP-1 drugs are linked to metabolic disease and weight control, the demographic differences are crucial. This shows that patient choices may differ greatly depending on the drug type.

2. Adverse Event Tolerability (Nausea): Nausea is the most reported and common gastrointestinal symptom of GLP-1 agonist medications' adverse events. The nausea rate is analyzed by proportion tests to confirm clinical expectations. The nausea rate among Ozempic users (19.67%) was consistent with the clinical control group, which is 20% (Wong et al., 2026). Furthermore, there was no statistically significant difference between the nausea rate of Ozempic users and Trulicity users. This indicates that gastrointestinal tolerability profiles are similar since these two brands share the same drug class, which is GLP-1 receptor agonists.

3. Clinical Complexity: Detailed knowledge about the variables affecting the overall number of reported adverse events was derived from the multiple linear regression and two-way ANOVA models. The results show that there is a strong positive correlation between clinical complexity, particularly polypharmacy (the number of concomitant drugs), and the reported adverse events count. Moreover, reporting behavior is strongly affected by biological sex. Female patient groups frequently reported a higher mean number of reactions than the male patient groups. Although the patients who use Mounjaro have the lowest mean

reaction counts and Ozempic has the highest, the important interaction effect revealed that the gender difference in reporting varies depending on the specific drug used.

Clinical Limitations

This study is based on reported cases of adverse events and patients under medical supervision registered in the healthcare system. The observed differences should be interpreted as variations in treatment rather than as direct evidence of the drug's effect. Real-world data is also significantly influenced by external factors such as prescribed dosages, individual dietary habits, duration of treatment, and overlapping side effects of concomitant medications.

Alignment of the Study with the Sustainable Development Goals (SDGs)

The findings revealed in this report, which combines complex health data with various statistical analyses, are directly related to the United Nations' global Sustainable Development Goals when viewed from a broader perspective.

- **SDG 3 (Good Health and Well-being):** Health systems, such as adverse event monitoring systems and patient tracking systems, shape the foundation of drug and patient safety. Understanding the real-life limits of drug tolerance and polypharmacy for all individuals in society supports healthcare professionals in making safer, evidence-based treatment choices for chronic metabolic diseases.
- **SDG 5 (Gender Equality):** When evaluating the general effects of medications on patients and adverse events, considering biological sex as a primary factor is one of the first steps toward a gender-sensitive health system. Documents showing that female and male patients report different reactions to different medications enable clinical safety profiles to be adapted according to specific gender-drug combinations and provide more equal healthcare for both genders.
- **SDG 10 (Reducing Inequalities):** Analyzing demographic factors, such as age and weight differences, among the groups of medication users identified in the study aims to identify critical differences in healthcare experiences that are vital to human life. Ensuring that real-world safety data considers various demographic variables plays a significant role in helping health systems identify sensitive patient subgroups, thereby reducing inequalities by making more inclusive and personalized healthcare services more accessible to everyone.

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